XFOIL: pressure, lift, camber, boundary layer

From the `xfoil` folder run `./xfoil`. Type one line at a time. [Enter] means press Enter on an empty line.

0. Load the airfoil

`NACA 0021`	load airfoil
`PANE`	160 panels
`OPER`	analysis menu — prompt `.OPERi` means inviscid

1. Inviscid

`CPMI -4`	C_p axis down to -4
`ALFA 10`	solve at $\alpha = 10^\circ \rightarrow C_p$ plot
`SEQP`	show the polar (not C_p) during sweeps
`PPAX`	polar axis limits (min max step):
`0 30 5`	α
`-0.5 4 0.5`	C_L
`0 0.25 0.05`	C_D
`-0.12 0.06 0.02`	C_M
`PACC`	start storing points (polar 1)
[Enter] [Enter]	no save file, no dump file
`ASEQ 0 30 2`	sweep $\alpha = 0, 2, \ldots, 30^\circ$
`PACC`	stop storing
`PPLO`	plot C_L vs α

2. Pressure on the airfoil surface

`ALFA 0`	
`CPV`	arrows out = suction ($C_p < 0$), arrows in = pressure ($C_p > 0$)
`ALFA 10`	
`CPV`	same, at 10°

3. Viscous

`VISC 5.8e6`	viscosity on, $Re = 5.8 \times 10^6$ (prompt `.OPERv`)
`ITER 200`	more iterations per point
`ALFA 0`	
`CPWR cp0021_a0.txt`	save C_p for part 4
`ALFA 10`	C_p: solid = viscous, dashed = inviscid
`CPWR cp0021_a10.txt`	save C_p for part 4
`PACC`	start storing (polar 2)
[Enter] [Enter]	
`ASEQ 0 30 2`	
`PACC`	
`PPLO`	red = inviscid, orange = viscous

4. Cambered: NACA 4421 (same 21% thickness, 4% camber)

[Enter]	back to top menu
`NACA 4421`	
`PANE`	
`OPER`	still viscous
`CREF`	overlay a saved C_p file (cyan symbols = 0021)
`ALFA 0`	asks for a filename →
`cp0021_a0.txt`	4421 lines vs 0021 symbols at 0°
`ALFA 10`	asks again →
`cp0021_a10.txt`	same, at 10°
`CREF`	overlay off (**must** be off before a sweep)
`PACC`	start storing (polar 3)
[Enter] [Enter]	
`ASEQ 0 30 2`	
`PACC`	
`PPAX`	zoom C_L axis:
[Enter]	α unchanged
`-0.5 2 0.5`	C_L
[Enter] [Enter]	C_D, C_M unchanged
`PPLO 2 3`	viscous 0021 (orange) vs 4421 (green)

5. Boundary layer: NACA 0021, $\alpha = 4°$

Transition x/c (T = top, B = bottom) and C_D are in the header of every `VPLO` plot.

(a) Natural transition

[Enter]	back to top menu
`NACA 0021`	
`PANE`	
`OPER`	still viscous, $N_{crit} = 9$, no trip
`ALFA 4`	
`BL 41`	41 velocity profiles drawn on the surface
`Z`	click two corners of a box around the trailing edge
`U`	unzoom (the zoom applies to every later plot)
`VPLO`	boundary-layer plot menu
`N`	amplification; transition where it reaches N_{crit} (dotted)
`H`	shape factor: about 2.6 laminar, about 1.4 turbulent
`CF`	skin friction: the jump is transition
`Y 0 0.01 0.002`	fix the C_f axis
`DUMP cf_natural.txt`	save this curve
[Enter]	back to OPER

(b) Trip at 5% chord

`VPAR`	viscous parameters
`XTR 0.05 0.05`	trip x/c on top, bottom
[Enter]	
`ALFA 4`	
`BL 41`	
`Z`	same box as (a): thicker boundary layer
`U`	
`VPLO`	
`CF`	
`Y 0 0.01 0.002`	
`OVER cf_natural.txt`	symbols = natural transition
[Enter]	

(c) No trip, change N_{crit}

9 = quiet tunnel, 4 = noisy

`VPAR`	
`XTR 1 1`	remove the trip
`N 4`	$N_{crit} = 4$
[Enter]	
`ALFA 4`	
`VPLO`	
`N`	
`CF`	
`Y 0 0.01 0.002`	
`OVER cf_natural.txt`	
[Enter]	then repeat (c) with `N 12`

If something goes wrong

[Enter] `/ QUIT`	[Enter] goes up one menu; `QUIT` exits from the top menu
`CPMI, PPAX`	curves run off a plot → widen its axis, then `CPX` or `PPLO`
`PLIS / PDEL 1`	list stored polars / delete polar 1 (extra ones appear every `PACC`)
`U`	undo a zoom
`HARD`	save the current plot to `plot.ps`